

Md Aman Alam

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PROFESSIONAL SUMMARY

Software Engineer at Capgemini and B.Tech Information Technology graduate focused on AI Engineering, Generative AI, and Cloud/DevOps, with hands-on experience building and deploying Agentic AI systems, RAG applications, and production-oriented ML services. Skilled in LangGraph, LangChain, FastAPI, PostgreSQL, Docker, AWS, CI/CD, LLM Evaluation, and Guardrails, with experience developing secure, tested, observable, and reliable AI applications.

SKILLS

Languages: Python, SQL, Java

AI/ML & GenAI: Scikit-learn, TensorFlow, XGBoost, LangChain, LangGraph, Hugging Face, LLMs, RAG, Embeddings, Vector Search, Prompt Engineering, Agentic AI, LLM Evaluation, Guardrails

Backend & Data: FastAPI, Streamlit, PostgreSQL, MongoDB, SQL Server, SQLite, ChromaDB, REST APIs

Cloud & MLOps: AWS, Microsoft Azure, Docker, Kubernetes, Terraform, Model Deployment, Model Serving, CI/CD Pipelines

DevOps & Engineering: Git, GitHub Actions, Jenkins, Linux, Pytest, Gitleaks, Trivy, Prometheus, Grafana

EDUCATION

Haldia Institute of Technology – B.Tech in Information Technology 2022–2026
CGPA: 7.96

D.A.V. Public School, Dhanbad 2021
12th: 81.33% — 10th: 85.3%

PROJECTS

AegisAI – Enterprise AI Operations Platform GitHub

- Built a production-style AI operations platform using FastAPI, Groq, PostgreSQL, Redis, and Docker, implementing JWT authentication, role-based authorization, rate limiting, caching, idempotency, retry, and circuit-breaker reliability controls.
- Implemented Prometheus metrics, OpenTelemetry tracing, audit logging, failure-injection testing, Streamlit operations dashboards, and GitHub Actions CI/CD for linting, testing, Docker validation, and image publishing.

Deepcite – Evidence-Grounded Multi-Agent Research GitHub

- Engineered a multi-agent research system using LangGraph, FastAPI, PostgreSQL, Groq, and Tavily MCP to coordinate planning, parallel research, evidence extraction, source verification, reasoning, and fact-checking.
- Implemented Human-in-the-Loop approval, resumable workflows, citation and confidence persistence, LangSmith observability, report evaluation, Docker Compose, and GitHub Actions CI for backend and frontend validation.

Agentic AI Customer Support System GitHub

- Architected a stateful Agentic AI customer support system using LangGraph with deterministic and LLM-based routing across specialized support workflows, Human-in-the-Loop escalation, and resumable execution using PostgreSQL checkpoint persistence.
- Built and deployed FastAPI and Streamlit services with 20 automated tests, Docker containerization, Neon PostgreSQL, and Render deployment, demonstrating end-to-end production-oriented AI system development.

EXPERIENCE

Capgemini – Cloud & DevOps Trainee (Left Shift Program) Jan 2026 – May 2026

- Provisioned cloud infrastructure using Terraform and CloudFormation, including VPCs, EC2 instances, networking components, and Security Groups.
- Automated CI/CD workflows using GitHub, Jenkins, Docker, Kubernetes, EKS, Minikube, and Helm.
- Implemented infrastructure and application monitoring using Prometheus and Grafana to improve system observability.
- Built a security-integrated DevSecOps CI/CD Pipeline using GitHub Actions, Gitleaks, Trivy, Infrastructure-as-Code security scanning, and automated Dev, Staging, and Production deployment workflows (GitHub).

Dataspace Academy – Machine Learning & Cloud Intern Jan 2025 – Mar 2025

- Developed and evaluated machine learning models using Python and Scikit-learn, applying data preprocessing, feature engineering, model training, evaluation, and cloud deployment practices across end-to-end ML workflows.